Comprehensive Formula Sheet: Probability & Stochastic Processes

Chapter 3: Distributions & Processes

Probability Distributions Table

Bernoulli:

$$P_X(x) = \begin{cases} 1-p, & x=0 \\ p, & x=1 \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = p, \quad \mathrm{Var}[X] = p(1-p)$$

Geometric:

$$P_X(x) = \begin{cases} p(1-p)^{x-1}, & x \geq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \frac{1}{p}, \quad \mathrm{Var}[X] = \frac{1-p}{p^2}$$

CDF of Geometric:

$$F_X(x) = \begin{cases} 0, & x < 1 \\ 1-(1-p)^{\lfloor x \rfloor}, & x \geq 1 \end{cases}$$

Binomial:

$$P_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, \ldots, n \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = np, \quad \mathrm{Var}[X] = np(1-p)$$

Pascal:

$$P_X(x) = \begin{cases} \binom{x-1}{k-1} p^k (1-p)^{x-k}, & x = k, k+1, \ldots \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \frac{k}{p}, \quad \mathrm{Var}[X] = \frac{k(1-p)}{p^2}$$

Poisson:

$$P_X(x) = \begin{cases} \dfrac{(\lambda T)^x e^{-\lambda T}}{x!}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \alpha, \quad \alpha = \lambda T, \quad \mathrm{Var}[X] = \alpha$$

Hypergeometric:

$$P_X(x) = \frac{\binom{r}{x}\binom{g}{n-x}}{\binom{r+g}{n}}, \quad E[X] = \frac{rn}{r+g}$$

Uniform (discrete):

$$P_X(x) = \begin{cases} \dfrac{1}{b-a+1}, & x = a, a+1, \ldots, b \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \frac{a+b}{2}, \quad \mathrm{Var}[X] = \frac{(b-a)(b-a+2)}{12}$$

Exponential:

$$f_X(x) = \begin{cases} ae^{-ax}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \frac{1}{a}, \quad \mathrm{Var}[X] = \frac{1}{a^2}$$

Gaussian:

$$f_X(x) = \begin{cases} \dfrac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, & \sigma > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \mu, \quad \mathrm{Var}[X] = \sigma^2$$

Uniform (continuous):

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \leq x < b \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \frac{a+b}{2}, \quad \mathrm{Var}[X] = \frac{(b-a)^2}{12}$$

Poisson Process and Related Formulae

Inter-arrival time of Poisson Process:

$$P[T_1 > t] = P[X(t) = 0] = e^{-\lambda t}$$

CDF of Exponential Distribution:

$$F_T(t) = P(T \leq t) = 1 - e^{-\lambda t}, \quad t \geq 0$$

n-th arrival time (Erlang-n distribution):

$$f_{T_n}(t) = \frac{\lambda^n t^{n-1}}{(n-1)!} e^{-\lambda t}, \quad t > 0, \ n = 1, 2, \ldots$$

Compound Poisson Process:

$$E\left(\sum_{i=1}^{N} X_i\right) = E(N)E(X)$$

$$\mathrm{Var}\left(\sum_{i=1}^{N} X_i\right) = E(N)\mathrm{Var}(X) + [E(X)]^2 \mathrm{Var}(N)$$

Superposition of Two Poisson Processes:

$$\lambda = \lambda_x + \lambda_y$$

Subdivision of Poisson Process:

$$\lambda_A = \lambda p, \qquad \lambda_B = \lambda(1-p)$$

Comparison of First Arrival Times:

$$P[T_X < T_Y] = \frac{\lambda_X}{\lambda_X + \lambda_Y}$$

Comparison of n-th and m-th Arrival Times:

$$P(S_n^1 < S_m^2) = \sum_{k=n}^{n+m-1} \binom{n+m-1}{k} \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n+m-1-k}$$

Merging of Bernoulli Process:

$$p = p_1 + p_2 - p_1 p_2$$

Conditional Probability and Laws

Probability of A, given B:

$$P[A|B] = \frac{P[AB]}{P[B]}, \quad P[B] > 0$$

Law of Multiplication (Product Rule):

$$P[AB] = P[A]P[B|A] = P[B]P[A|B]$$

Chain Rule:

$$P[ABC] = P[A]P[B|A]P[C|AB]$$

$$P[A_1 A_2 \ldots A_n] = P[A_1]P[A_2|A_1]P[A_3|A_1A_2] \ldots P[A_n|A_1A_2 \ldots A_{n-1}]$$

$$= \prod_{i=1}^{n} P\left(A_i \,\middle|\, \bigcap_{j=1}^{i-1} A_j\right)$$

Law of Total Probability

Let B be an event with $P[B] \geq 0$ and $P[B^c] \geq 0$, then:

$$P[A] = P[A|B]P[B] + P[A|B^c]P[B^c]$$

Let $B = \{B_1, B_2, \ldots, B_n\}$ be an event space:

$$\begin{aligned} P[A] &= P[A|B_1]P[B_1] + P[A|B_2]P[B_2] + \cdots + P[A|B_n]P[B_n] \\ &= \sum_{i=1}^{n} P[A|B_i]P[B_i] \end{aligned}$$

Expected Value and Moments

Expected Value:

$$E[X] = \sum_x x\, P_X(x) = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

Moments:

$$E[X^n] = \sum_x x^n P_X(x) = \int_{-\infty}^{\infty} x^n f_X(x)\, dx$$

Central Moments:

$$E[(X-\mu_X)^n] = \sum_x (x-\mu_X)^n P_X(x) = \int_{-\infty}^{\infty} (x-\mu_X)^n f_X(x)\, dx$$

Covariance and Correlation

$$\begin{aligned} C_{XY} = \mathrm{Cov}[X,Y] = \sigma_{XY} &= E[(X-\mu_X)(Y-\mu_Y)] \\ &= E[XY] - \mu_X\mu_Y \end{aligned}$$

$$\rho_{XY} = \frac{C_{XY}}{\sigma_X \sigma_Y}, \quad -1 \leq \rho \leq 1$$

Conditional Expectation

$$E[X|A] = \sum_x x P_{X|A}(x) = \int_{-\infty}^{\infty} x f_{X|A}(x)\, dx$$

$$E[X|Y=y] = \sum_x x P_{X|Y}(x|y) = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\, dx$$

Law of Total Expectation:

$$E[X] = \sum_y E[X|Y=y]P_Y(y) = \int_{-\infty}^{\infty} E[X|Y=y] f_Y(y)\, dy$$

Chapter 4: Discrete-Time Markov Chains

Fundamental Properties

Markov Property:

$$P[X_k = j \mid X_{k-1} = i, \ldots] = P[X_k = j \mid X_{k-1} = i]$$

Homogeneous Chain (Time Independent):

$$p_{ij} = P[X_k = j \mid X_{k-1} = i]$$

Joint Probability (Markov Chain Rule):

$$P[X_k = j, \ldots, X_0 = i_k] = p_{i_1 j} \ldots p_{i_k i_{k-1}} P[X_0 = i_k]$$

n-Step Transition Probability:

$$p_{ij}(n) = P[X_{m+n} = j \mid X_m = i]$$

Chapman-Kolmogorov Equations:

$$p_{ij}(n) = \sum_k p_{ik}(r) p_{kj}(n-r), \quad 0 < r < n$$

Matrix Form:

$$P^{(n)} = P^n$$

First Passage and Recurrence

First Passage Probability $f_{ij}(n)$:

$$f_{ij}(n) = \sum_{l \neq j} p_{il} f_{lj}(n-1)$$

Total probability of ever reaching j:

$$f_{ij} = \sum_{n=1}^{\infty} f_{ij}(n)$$

Mean First Passage m_{ij} $(i \neq j)$:

$$m_{ij} = 1 + \sum_{k \neq j} p_{ik} m_{kj}$$

Mean Recurrence m_{ii}:

$$m_{ii} = 1 + \sum_{k \neq i} p_{ik} m_{ki}$$

Steady-State Distribution

Limiting Probabilities π:

$$\pi_j = \lim_{n \to \infty} p_{ij}(n)$$

System of Equations:

$$\pi_j = \sum_k \pi_k p_{kj} \quad (\text{or } \boldsymbol{\pi} = \boldsymbol{\pi} P)$$

$$1 = \sum_j \pi_j$$

Sojourn Time

PMF:

$$p_{D_i}(d) = p_{ii}^{d-1}(1-p_{ii})$$

Mean:

$$E[D_i] = \frac{1}{1-p_{ii}}$$

Transient Analysis

z-Transform:

$$G(z) = [I - Pz]^{-1} = \sum_{n=0}^{\infty} P^n z^n$$

Decomposition:

$$P^n = C + T(n)$$

Mean Occupancy $\phi_{ik}(n)$:

$$\Phi(n) = \sum_{r=0}^{n} P^r$$

Absorbing Chains

Canonical Form:

$$P = \begin{bmatrix} I & \mathbf{0} \\ R & Q \end{bmatrix}$$

Fundamental Matrix N:

$$N = [I - Q]^{-1} = \sum_{k=0}^{\infty} Q^k$$

Mean Time to Absorption μ_i:

Vector form:

$$M = N\mathbf{1} = [I - Q]^{-1}\mathbf{1}$$

Recursive form:

$$\mu_i = 1 + \sum_{j \in T} p_{ij}\mu_j$$

Absorption Probabilities B:

$$B = NR = [I - Q]^{-1} R$$

Reversible Chains

Detailed Balance Condition:

$$\pi_i p_{ij} = \pi_j p_{ji}$$

Reversed Transition Probabilities:

$$\hat{p}_{ij} = \frac{\pi_j p_{ji}}{\pi_i}$$

Chapter 5: Continuous-Time Markov Chains

Fundamental Definitions

Markov Property:

$$P[X(t+s) = j | X(s) = i, X(u) = k, 0 \le u \le s]$$
$$= P[X(t+s) = j | X(s) = i]$$

Transition Probabilities:

$$p_{ij}(t) = P[X(t+s) = j | X(s) = i]$$
$$p_j(t) = P[X(t) = j]$$
$$\sum_j p_{ij}(t) = 1, \quad \sum_j p_j(t) = 1$$

Chapman-Kolmogorov Equation:

$$p_{ij}(t+s) = \sum_k p_{ik}(t) p_{kj}(s)$$

$$P(t+s) = P(t)P(s)$$

Holding Time and Transition Rates

Holding Time Distribution:

$$T_i \sim \exp(\nu_i)$$

Transition rate: ν_i
Expected holding time:

$$E[T_i] = \frac{1}{\nu_i}$$

State Probability at $t + \Delta t$:

$$P_i(t+\Delta t) = P_i(t) \left\{ 1 - \sum_{j \ne i} P_{ij} \nu_i \Delta t \right\} + \sum_{j \ne i} P_j(t) P_{ji} \nu_j \Delta t$$

Steady State Probability:

$$0 = -P_i(t)\nu_i \sum_{j \ne i} P_{ij} + \sum_{j \ne i} P_j(t) P_{ji} \nu_j$$

with $\sum_i P_i = 1$
Rate Equation:

$$P_i \sum_{j \ne i} \nu_i P_{ij} = P_i \sum_{j \ne i} \nu_{ij}$$

Infinitesimal Generator Matrix

Transition Rates:

$$q_{ji} = p_{ji}\nu_j, \quad q_i = \nu_i \sum_{j \ne i} p_{ij} = \sum_{j \ne i} q_{ij}$$

Generator Matrix Q:

$$Q = \begin{bmatrix} -q_1 & q_{12} & q_{13} & \cdots \\ q_{21} & -q_2 & q_{23} & \cdots \\ q_{31} & q_{32} & -q_3 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Transient Analysis

Differential Equation:

$$\frac{dp_i(t)}{dt} = -q_i p_i(t) + \sum_{j \ne i} p_j(t) q_{ji}$$

Matrix Form:

$$\frac{dp(t)}{dt} = p(t)Q$$

Solution (with $p(0) = I$):

$$p(t) = e^{Qt} = I + \sum_{k=1}^{\infty} \frac{Q^k t^k}{k!}$$

Kolmogorov Forward Equations:

$$\frac{dp_{ij}(t)}{dt} = \sum_{k \ne j} p_{ik}(t) q_{kj} - p_{ij}(t) q_j$$

$$\frac{dp_{ij}(t)}{dt} = \sum_k p_{ik}(t) q_{kj}$$

Steady-State Analysis

Limiting Probabilities:

$$\lim_{t \to \infty} p(t) = p, \quad \lim_{t \to \infty} \frac{dp(t)}{dt} = 0$$

Balance Equations:

$$pQ = 0$$
$$pe^T = 1, \quad e^T = [1, 1, 1, \ldots]^T$$

Birth and Death Processes

Transition Rates:

$$\lambda_i = \nu_i p_{i(i+1)} \quad \text{(birth rate)}$$

$$\mu_i = \nu_i p_{i(i-1)} \quad \text{(death rate)}$$

$$\mu_0 = 0$$

Differential Equations:

$$\frac{dp_0(t)}{dt} = -\lambda_0 p_0(t) + \mu_1 p_1(t)$$

$$\frac{dp_i(t)}{dt} = -(\lambda_i + \mu_i) p_i(t) + \mu_{i+1} p_{i+1}(t) + \lambda_{i-1} p_{i-1}(t)$$

Steady-State Balance:

$$\lambda_0 p_0 = \mu_1 p_1$$

$$(\lambda_i + \mu_i) p_i = \mu_{i+1} p_{i+1} + \lambda_{i-1} p_{i-1}, \quad i \ge 1$$

$$\sum_i p_i = 1$$

Local Balance:

$$\lambda_i p_i = \mu_{i+1} p_{i+1}, \quad i = 0, 1, 2, \ldots$$

Recursive Solution:

$$p_{i+1} = \frac{\lambda_i}{\mu_{i+1}} p_i = \frac{\lambda_i \lambda_{i-1} \cdots \lambda_0}{\mu_{i+1} \mu_i \cdots \mu_1} p_0$$

$$p_0 = \left[1 + \sum_{i=1}^{\infty} \frac{\lambda_{i-1} \lambda_{i-2} \cdots \lambda_0}{\mu_i \mu_{i-1} \cdots \mu_1} \right]^{-1}$$

$$p_i = \frac{\lambda_{i-1} \lambda_{i-2} \cdots \lambda_0}{\mu_i \mu_{i-1} \cdots \mu_1} p_0, \quad i \ge 1$$

Special Case ($\lambda_i = \lambda$, $\mu_i = \mu$):
Converges if $\lambda < \mu$:

$$p_0 = 1 - \frac{\lambda}{\mu}$$

$$p_i = \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^i, \quad i \ge 1$$

First Passage Time

Mean First Passage Time:

$$T_k = \min\{t \geq 0 | X(t) = k\}$$

$$m_{ik} = E[T_k | X(0) = i]$$

Equation:

$$\nu_i m_{ik} = 1 + \sum_{j \neq k} \nu_{ij} m_{jk}$$

Alternative Form:

$$m_{ik} = \frac{1}{\nu_i} + \sum_{j \neq k} p_{ij} m_{jk}$$

where $p_{ij} = \nu_{ij}/\nu_i$.

Chapter 6: Queueing Theory

M/M/1/K

$$\lambda_n = \begin{cases} \lambda, & n < K \\ 0, & n \geq K \end{cases}$$

$$p_0 = \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^{K+1}}$$

$$p_n = \left(\frac{\lambda}{\mu}\right)^n p_0, \quad 0 \leq n \leq K$$

$$E[N] = \frac{\rho}{1-\rho} - \frac{(K+1)\rho^{K+1}}{1-\rho^{K+1}}$$

$$E[N_q] = E[N] - (1 - p_0)$$

Customer turned away with prob. p_k

$$p_K = \frac{(1-\rho)\rho^K}{1-\rho^{K+1}}$$

Actual arrival rate λ_A

$$\lambda_A = \lambda(1 - p_K)$$

$$E[T] = \frac{E[N]}{\lambda_A}, \quad E[W] = \frac{E[N_q]}{\lambda_A}$$

General Birth–Death

$$p_0 = \frac{1}{1 + \sum_{n=1}^{N} \frac{\lambda_{n-1} \cdots \lambda_0}{\mu_n \cdots \mu_1}}$$

$$p_n = \frac{\lambda_{n-1} \cdots \lambda_0}{\mu_n \cdots \mu_1} p_0$$

Little's Law

$$N = \lambda T$$

$$\lambda_t = \frac{a(t)}{t}, \quad T_t = \frac{\gamma(t)}{a(t)}, \quad \bar{N}_t = \frac{\gamma(t)}{t}$$

$$E[N] = \lambda E[T]$$
$$E[N_q] = \lambda E[W]$$

M/M/1

$$p_0 = 1 - \frac{\lambda}{\mu}$$

$$p_n = \left(\frac{\lambda}{\mu}\right)^n p_0$$

$$\rho = \frac{\lambda}{\mu}$$

$$E[N] = \frac{\rho}{1-\rho}$$

$$E[T] = \frac{E[N]}{\lambda}$$

$$E[W] = \frac{\rho}{\mu(1-\rho)}$$

$$E[N_q] = \lambda E[W] = \frac{\rho^2}{1-\rho}$$

$$E[N_s] = E[N] - E[N_q] = \rho$$

M/M/∞

$$\mu_n = n\mu$$

$$p_0 = \left(\sum_{n=0}^{\infty} \frac{(\lambda/\mu)^n}{n!}\right)^{-1} = e^{-\lambda/\mu}$$

$$p_n = \frac{(\lambda/\mu)^n}{n!} p_0$$

$$E[N] = \frac{\lambda}{\mu}, \quad T = \frac{1}{\mu}$$

M/M/m or M/M/c

$$\mu_n = \begin{cases} n\mu, & 0 < n < m \\ m\mu, & n \geq m \end{cases}$$

$$\lambda p_{n-1} = \min(n, c)\, \mu\, p_n, \qquad n = 1, 2, \ldots$$

$$p_0 = \left[\sum_{n=0}^{c-1} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n + \frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c \frac{c\mu}{c\mu - \lambda}\right]^{-1}, \qquad \lambda < c\mu$$

$$p_n = \begin{cases} \frac{(\lambda/\mu)^n}{n!} p_0, & n < m \\ \frac{(\lambda/\mu)^n}{m!\, m^{n-m}} p_0, & n \geq m \end{cases}$$

$$p_c = \frac{(\lambda/\mu)^c}{c!} p_0$$

$$p_w = \frac{p_c}{1 - \frac{\lambda}{c\mu}} = \frac{p_c}{1-\rho}$$

$$E[N_q] = \frac{\rho\, p_c}{(1-\rho)^2}$$

$$E[W] = \frac{E[N_q]}{\lambda} = \frac{p_c}{c\mu(1-\rho)^2} = \frac{P_W}{c\mu(1-\rho)}$$

M/M/m/m

$$\lambda_n = \lambda, \quad n < m. \qquad \mu_n = n\mu$$

$$p_0 = \left(\sum_{k=0}^{m} \frac{(\lambda/\mu)^k}{k!}\right)^{-1}$$

$$p_n = \frac{(\lambda/\mu)^n}{n!} p_0, \qquad \lambda_A = \lambda(1 - p_c)$$

$$E[N] = \frac{\lambda(1-p_c)}{\mu}, \quad E[T] = \frac{E[N]}{\lambda_A}$$

M/M/1/ /K

$$\lambda_n = \begin{cases} (K-n)\lambda, & 0 \leq n < K, \\ 0, & n = K. \end{cases}$$

$$\mu_n = \mu, \qquad n \geq 1.$$

Local Balance Equations $(K-n+1)\lambda\, p_{n-1} = \mu\, p_n$

Steady-State Probabilities

$$p_n = \frac{K!}{(K-n)!}\left(\frac{\lambda}{\mu}\right)^n p_0, \qquad n = 0, 1, 2, \ldots, K.$$

$$p_0 = \left[\sum_{n=0}^{K} \frac{K!}{(K-n)!}\left(\frac{\lambda}{\mu}\right)^n\right]^{-1}.$$

$$p_n = \frac{\dfrac{K!}{(K-n)!}\left(\dfrac{\lambda}{\mu}\right)^n}{\displaystyle\sum_{j=0}^{K} \frac{K!}{(K-j)!}\left(\frac{\lambda}{\mu}\right)^j}, \qquad n = 0, 1, \ldots, K.$$